MILITARY SPECIFICATION

TANK, STORAGE, LIQUID OXYGEN TMU-7A/E

This amendment forms a part of Military Specification MIL-T-27892F dated 15 Dec 1967 and is approved for use by the Department of the Air Force and is available for use by all Departments and agencies of the Department of Defense.

Page 1

*2.1 Under Specifications, Delete the following:

Federal

O-T-634, HH-I-551, and PPP-T-60.

Military

MIL-T-5021, MIL-I-6866 and MIL-I-6868.

*2.1 Under Specifications, Add the following:

Military

MIL-C-81302	Cleaning Compound Solvent, Trichlorotrifluoroethane.
MIL-C-83286	Coating Urethane, Aliphatic Isocyanate, for Aerospace Applications.

*2.1 Under Standards, Delete the following:

MIL-STD-100, MIL-STD-1186, and MS33586.

*2.1 Under Standards, Add the following:

Federal

FED-STD-595	Colors.

Military

DOD-STD-100	Engineering Drawing Practices.
MIL-STD-889	Dissimilar Metals.
MIL-STD-1359	Cleaning Methods and Procedures for Breathing Oxygen Equipment.
MIL-STD-1595	Qualification of Aircraft, Missile and Aerospace Fusion Welders.

FSC 3655

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*2.1 Change the Titles on the following documents as indicated:

L-S-300	Sheeting and Tape, Reflective, Nonexposed Lens.
MIL-V-173	Varnish, Moisture, and Fungus Resistant (For Treatment of Communications, Electronic and Associated Equipment).
MIL-R-3065	Rubber, Fabricated Products.
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of.
MIL-STD-808	Finishes, Materials and Processes for Corrosion Prevention and Control in Support Equipment.

Page 2

*Under Drawings, Add the following:

Ordnance Corps

C8987855	Seat, LOX Coupling 1".
C8987869	LOX Cap 1".

Air Force

8221044-01	Decal

*2.2 Other Publications, Add the following:

American Society for Testing and Materials (ASTM).

ASTM D 3951	Standard Practice for Commercial Packing.

Page 3

*3.4.2.2 Dissimilar Metals, Change the last sentence of this paragraph to read, Dissimilar Metals are defined in MIL-STD-889.

3.4.3 Gasketing and Insulating Materials, Add the following sentence to the end: Mylar super insulation or perlite are acceptable for annular insulation space.

3.4.6 Materials, Add the following paragraph under Section 3.4 Materials:
3.4.6 Recycled, Virgin, and Reclaimed Materials, Provided that all other requirements of this specification are met, reclaimed materials shall be used to the maximum extent possible with no exclusion to the use of recovered materials and no requirement that an item be manufactured from virgin materials.

*3.5.6.1 Aircraft Loading and Transport, Change the first sentence to read, The tank design and configuration shall permit loading into and air transport by Types C-5, C-141, and C-130 Aircraft.

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*3.5.6.2 Delete and substitute, Equipment Restraint Criteria, The tank shall withstand the equipment restraint criteria forces specified in MIL-A-8421 when transported empty.

*3.6.4 Pressure Vessels, Change to read, Unless otherwise specified herein, the liquid storage tank assembly, piping, and connections thereto, shall be fabricated in accordance with Section VIII, Division 1, of ASME Boiler and Pressure Vessel Code. The pressure vessel shall be cylindrical, and shall be mounted to the skid base such that its longitudinal axis is horizontal and shall meet the dimensional requirements of paragraph 3.10.

Page 7

3.6.11 Certification of Welders, Change MIL-T-5021 to ASME Code, Sect. IX.

Page 8

3.7.3 Change "100 Microns" to "35 Microns".

3.7.3a Change "100 Microns" to "35 Microns".

*3.7.4 Air Transportability, Change to read, The tank shall not be damaged by the equipment restraint criteria specified in MIL-A-8421 when transported empty.

*3.7.4.1 Change to read, The empty tank shall also withstand, without damage, like forces as imposed by the equipment restraint criteria specified in MIL-A-8421 while being transported on a railroad car, seagoing vessel, or truck.

3.8.1.1e Delete Manway.

*3.8.1.2 Delete and substitute, Fabrication, the liquid storage tank assembly shall be a welded vessel designed and fabricated in accordance with Section VIII, Div 1, of the ASME Boiler and Pressure Vessel Code to contain and retard evaporation loss of liquid oxygen as necessary to insure compliance with the performance requirements specified in 3.7.1 and 3.7.2. Only the inner shell shall be stamped per ASME Code, Section VIII, Division 1, for the specified maximum allowable working pressure.

3.8.1.2.1 Delete and substitute, Component Arrangement, The liquid storage tank assembly shall be fabricated with the inner shell suspended within the outer shell to maintain a vacuum-tight annular insulation space of sufficient distance between corresponding points on the shells and heads to meet the requirements specified herein.

Page 9

3.8.1.3 Delete and substitute, Inner Shell, The inner shell shall physically store the liquid oxygen. Insofar as practicable, it shall be designed for 100% joint efficiencies and 100% radiography of longitudinal and circumferential welded joints per ASME Code, Section VIII, Division 1, para UW-11a. The minimum surface area shall be consistent with the requirements specified herein.

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*3.8.1.3.2 Change to read, Inner Shell Material, The inner shell shall be fabricated from 18CR-8NI Stainless Steel Material Type 304 to insure compliance with the requirements specified herein, and that is resistant to corrosion caused by contact with liquid or gaseous oxygen, nitrogen, moist air, water, or cleaning solvents such as trichlorotrifluoroethane.

*3.8.1.3.3 Delete and substitute, Inner Shell Design Pressure, (see 6.3.5) The inner shell shall be designed and constructed for a maximum allowable working pressure (55 psig) adequate to permit the vapor space pressure relief valve setting specified in 3.8.1.9.3.1.1 and for a normal working pressure of not less than (50 psig) for transferring liquid oxygen under the conditions specified herein. Design and construction shall also be such as to insure against:

a. Distortion or damage when the liquid storage tank assembly annular insulation space is evacuated to a full vacuum (15 psia) or is pressurized to the outer shell pressure relief device maximum opening pressure.

b. Collapse of, or damage to the inner shell when it is evacuated to an absolute pressure of 100 microns mercury (HG) or less with the annular insulation space at 15 psia pressure.

*3.8.1.4 Delete and substitute, Outer Shell, The outer shell shall provide support for the inner shell and inner shell suspension system. It shall be designed for 70% joint efficiencies and NO radiography of longitudinal or circumferential welded joints is required per ASME Code, Section VIII, Div 1, para UW-11c. The underside of the outer shell shall be provided with reinforced pads for the saddles support and to comply with the requirements specified herein and shall be equipped with skids specified in 3.8.5 for mounting the tank on a flat concrete pad or other suitable surface.

Page 10

3.8.1.4.1 Outer Shell Material, Change to read, The outer shell shall be fabricated from material SA-515 GR 60, or GR 65; or SA-516 GR 60 or GR 65 which has sufficient strength, rigidity, and vacuum holding properties to insure compliance with design temperature of -300°F to +150°F and with the requirements specified herein.

*3.8.1.4.2 Delete and substitute, Outer Shell Design Pressure, The outer shell shall be designed and constructed in accordance with Section VIII, Div 1, of the ASME Boiler and Pressure Vessel Code for full vacuum (15 psia) to insure sufficient strength and rigidity to prevent:

a. Distortion or damage when the liquid storage tank assembly annular insulation space is evacuated to a full vacuum or pressurized to the outer shell pressure relief device maximum opening pressure.

b. Collapse of, or damage to the outer shell during performance of maintenance functions.

*3.8.1.4.2.1 Add a new paragraph Outer Shell Support of Inner Shell, The outer shell shall be designed and constructed in accordance with Section VIII of the ASME Boiler and Pressure Vessel Code in addition to the require-

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ments of 3.8.1.4.2. The outer shell shall be designed to support the weight and contents of the inner shell in a manner which will meet the requirements of 3.7.2 and 3.7.3.

3.8.1.4.3.2 Change the first sentence to read, If a powder type insulation is used, the vacuum line filter shall be a removable flanged and bolted type unit extending into the tank annular space and shall have an effective filtering area of not less than 500 square inches.

Page 11

3.8.1.7 Delete.

3.8.1.7.1 Delete.

3.8.1.7.2 Delete.

Page 12

3.8.1.7.3 Delete.

3.8.1.7.4 Delete.

3.8.1.7.5 Delete.

3.8.1.8.1.1.1.1e Delete.

Page 13

*3.8.1.8.1.4.1 Delete and substitute, Liquid Fill Drain Line Filter Design, The filter design shall be similar to Air Force Drawing 8321391, except that it shall be connected at both ends to the 2-inch liquid fill drain line, and shall have a flow rate of 100 gpm at an operating pressure of 250 psig with a pressure drop across the filter not to exceed 3 psig. Operating temperature range shall be -350°F to +250°F.

Page 14

Figure 2, Delete entirely.

Page 15

3.8.1.8.1.4.2 Delete entirely.

Table 1, Delete entirely.

Page 16

3.8.1.8.1.5.3 Servicing Line Disconnect Fitting, Add the following sentence to the end, A one inch male liquid oxygen coupling half assembly consisting

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of a seat, cap, and a gasket conforming to Drawings C8987855, C8987869, B88987832 respectively, assembled as shown on Drawing C8987830 shall be provided on the one inch external pipe thread.

3.8.1.8.1.5.4 Servicing Line Filter, Change the last sentence to read, The filter design shall be in accordance with Air Force Drawing 8321391.

3.8.1.8.1.5.4.1 Delete entirely.

3.8.1.8.1.5.5 Delete entirely.

Page 18

3.8.1.9 Safety Devices, Change the last sentence to read, Cleaning solvents such as trichlorotrifluoroethane, per MIL-C-81302.

Page 21

3.8.1.10.1.2 Valve Material, Change the last word in this paragraph from trichloroethylene to trichlorotrifluoroethane, per MIL-C-81302.

Page 23

3.8.1.11.3 Delete and substitute, Vacuum Sensing Tap, A thermocouple gage tube National Stock Number 6685-00-877-9593 shall be installed in the vacuum system. The tube shall be mounted with a vacuum-tight isolation valve.

*3.8.1.11.3.1 Delete entirely.

3.8.1.11.3.2 Vacuum Connection, Add the following sentence to the end: Use thermocouple valve part number 4551F28 (MFC 28968) or equivalent.

3.8.1.11.3.3 Delete entirely.

*3.8.2.1 Delete and substitute, Buildup Rate, After calculated adjustment to standard temperature and pressure (STP) the liquid storage tank pressure buildup system shall pressurize the inner shell vapor space to 30 psig within 10 minutes after closing the vapor vent line shutoff valve and initiating buildup when the innershell is filled to between 15 and 20 percent of rated capacity and has been vented to atmospheric pressure for not less than four (4) hours.

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3.8.3.5 Liquid Servicing Hose Storage Provisions, Change "the acceleration forces specified in MIL-A-8421", to read, "equipment restraint criteria specified in MIL-A-8421".

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*3.10 Dimensions, Line 8, Delete "9" and substitute "8".

3.12.1 Surfaces contacting oxygen, change trichloroethylene to trichlorotrifluoroethane, per MIL-C-81302.

3.12.2.1 Delete and substitute, Finish Code F-105, shall apply to all exterior parts and surfaces of the tank, tank color shall be Glass White No 17875 or equivalent per MIL-C-83286 unless otherwise noted on Acquisition Data Sheet Form 2.

3.12.2.2 Add to the end of the sentence, per MIL-C-83286.

*3.13.1 Delete and substitute, Tank Marking, Tank description and warning markings shall be in the form of decalcomanias which shall be government furnished property. Each tank shall require six each of the decalcomania described by San Antonio Air Logistics Center, SFRM Sketch 8221044-01 for Liquid Oxygen. The decalcomanias shall be applied 4 each on the sides of the tank; that is, 2 on each side equidistant from each other and from their respective ends of the tank, and one each centered on each end of the tank.

3.13.1.1 Delete entirely.

3.13.1.2 Delete entirely.

Page 30

3.14.1 Tank Nameplate Location, Add the following to the end: Tank nameplate part number 44078 (MFC 98750, NSN-3655-01-064-0894YD) shall be furnished by SA-ALC/SFSC.

*3.16.1 Delete and substitute, Degreasing, Tank surfaces, parts, fittings, etc. that will be in contact with high purity oxygen shall be degreased in accordance with procedures in MIL-STD-1359. Precautions shall be taken to insure that solvents do not contact parts fabricated from incompatible materials.

Page 31

3.17 Delete entirely.

*4.1 Add the following sentence to the end of the paragraph: No tests or inspections shall be performed until the test plan has been approved by the responsible government representative and only after the government representative has approved of the time and location of the tests/inspections.

4.2(a,) Delete preproduction testing and substitute first article testing.

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4.3.2.1 After "servicing line", change to read, "shall be operated by the gas pressure and measured by differential pressure meters furnished as part of the tank".

4.3.2.1.1 Line four, change "gages" to read "gases".

4.3.5 Change "100 microns" to "35 microns".

Page 34

*4.4 Delete and substitute, First Article Inspection, shall consist of the examination and tests specified in 4.6.

*4.4.1 Delete and substitute, First Article Sample, Unless otherwise specified, as soon as practicable after the award of the contract or purchase order, the contractor shall submit one tank. The sample shall be representative of the construction, workmanship, components, and materials to be used during production. When a contractor is in continuous production of this tank, from contract to contract, submission of further first article inspection samples, on the new contract, may be waived at the discretion of the procuring activity (see 6.2). Approval of the first article inspection sample or the waiver of the first article inspection does not preclude the requirements for performing the quality conformance inspection. The first article sample shall be plainly identified by a securely attached durable tag that shall be marked with the following information:

Sample submitted by (name), (date), for first article inspection in accordance with MIL-T-27892 under Contract No.

4.4.2 Line 1, Delete "the preproduction" and substitute "first article".

Page 36

*4.5.2.3 Delete and substitute, Sampling Tests of Filters, The liquid fill drain line and the servicing line filters shall be run through the Acceptance Tests of Air Force Drawing 8321391.

4.5.2.3.1 Delete entirely.

4.5.2.3.2 Delete entirely.

4.5.2.3.2.1 Delete entirely.

Page 38

*4.6.5.3 Delete and substitute,
Equipment Restraint Criteria, capabilities of the tank design and construction fulfilling the equipment restraint criteria specified in MIL-A-8421 and shall be included as part of the test report specified in 4.4.2. Calculation shall be approved by the engineer in the procuring activity before the performance of the preproduction testing paragraph 4.4.

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4.6.6.1.1 Change "100 microns" to "35 microns".

Page 44

4.6.15 Delete and substitute, Filter Tests, per Air Force Drawing 8321391.

4.6.15.1 Delete entirely.

4.6.15.2 Delete entirely.

Page 45

4.6.15.2.1 Delete entirely.

4.6.15.3 Delete entirely.

4.6.15.3.1 Delete entirely.

4.6.15.4 Delete entirely.

Page 46

4.6.18.1 Change the last sentence to read: Magnetic particle inspection shall be performed in accordance with ASME Code Section V.

4.6.18.2 Change the last sentence to read, Dye penetrant inspection shall be performed in accordance with ASME Code Section V.

Page 48

*5.2 Physical Protection, Delete MIL-STD-1186 and replace with ASTM D 3951.

5.4 Delete entirely.

5.4.1 Delete entirely.

*6.2 Ordering Data.

6.2.1 Acquisition Requirements, Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a complete set of all special tools, other than common hand-tools and those normally available in motor vehicle repair shops, are to be provided. (see 3.8.6).

c. Whether first article inspection is waived (see 4.4.1).

d. Selection of applicable levels of preservation and packing required (see 5.2 and 5.3).

e. When tank is not to be provided with a wooden shield (see 5.3.1).

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*6.2.2 First Article, When a first article inspection is required, the tank will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangement for examinations, tests, and approval of the first article.

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6.3.7 Delete entirely.

*6.4 Delete and substitute, Changes from Previous Issue, The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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